A Walk Through Calculus of Variations

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Methods in Applied Mathematics

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Outline



Background

Example: Minimizing a Functional

Solving Partial Differential Equations

An Example Using Beltrami Identity

What is Calculus of Variations?



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- Some well known problems: brachistochrone curve, Newton's minimal resistance problem...

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- Consider the $L^p(\mathbb{R})$ norm, that is

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then $I[\cdot] := \|\cdot\|_{L^p(\mathbb{R})} = \left(\int_{\mathbb{R}} |\cdot|^p \, dx \right)^{1/p}$ is a functional that sends $f \mapsto I[f] \in \mathbb{R}$ a real number.

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- Euclidean Arclength functional

$$f \mapsto L[f] := \int_{t_0}^{t_1} \sqrt{1 + \left(\frac{df}{dx} \right)^2} \, dx \in \mathbb{R}$$

Minimizing a functional

- Given the functional $J[\cdot] : y \mapsto \mathbb{R}$ where

$$J[y] := \int_0^1 1 + (y')^2 \, dx$$

with boundary conditions $y(0) = 0$ and $y(1) = 1$.

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- Typical calculus method would tell us to solve the extremum problem

$$\frac{dJ}{dy} = 0 \Longrightarrow y = ?$$

  but what does it mean to differentiate with respect to a function?

Minimizing a functional

This is where we use a *variational approach*. Let h be another function and let $\epsilon > 0$ small, then $y + \epsilon h$ is a small perturbation from y. Now we compute the limit

$$\lim_{\epsilon \to 0} \frac{J[y + \epsilon h] - J[y]}{\epsilon} = \lim_{\epsilon \to 0} \frac{\displaystyle\int_0^1 1 + (y' + \epsilon h')^2 - (1 + (y')^2)\, dx}{\epsilon}$$

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$$= \lim_{\epsilon \to 0} \int_0^1 2y' h' + \epsilon (h')^2\ dx = \int_0^1 2y' h'\ dx$$

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Therefore, we have

$$\int_0^1 2y''h \; dx = 0 \text{ for all } h \in C^1[0, 1].$$

Minimizing a functional

Therefore $y''(x) = 0$, if we integrate twice with respect to x we get

$$y(x) = C_1 x + C_2$$

which is the equation of a straight line. If we utilize our boundary values we have

$$y(x) = x.$$

Solving PDE's

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- There is no general solution for solving such a problem.
- If we consider the particular class of problems where $A[\cdot] = J'[\cdot]$ where A is the derivative of a functional J, also called the energy. Then our problem becomes

$$J'[u] = 0$$

Again we are finding the critical values, "functions", that solve the above extremum problem.

The First Variation

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$$L = L(p, z, x), \quad p \in \mathbb{R}^n, z \in \mathbb{R}, x \in \Omega$$

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Note, when we take gradients we have

$$\begin{cases} D_p L = (L_{p_1}, \ldots, L_{p_n}) \\ D_z L = L_z \\ D_x L = (L_{x_1}, \ldots, L_{x_n}) \end{cases}$$

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Now suppose

$$J[w] := \int_\Omega L(Dw(x), w(x), x)\ dx$$

for smooth functions $w \in \mathcal{W} = \{w : \overline{\Omega} \to \mathbb{R}, w = g \text{ on } \partial\Omega\}$

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- Let $\epsilon \to 0$ to get

$$j'(0) = \eta J'[u] = 0;$$

therefore $j(\cdot)$ has a minimum at $\epsilon = 0$.

The First Variation

Now let us explicitly compute $j'(\epsilon)$, the *first variation*:

$$j'(\epsilon) = \frac{d}{d\epsilon} \int_\Omega L(Du + \epsilon D\eta, u + \epsilon\eta, x)\ dx$$

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For the first integral we can apply integration by parts to each term. That is let $U = L_{p_k}(Du, u, x)$, then $dU = (L_{p_k}(Du, u, x))_{x_k} \, dx$, and let $dV = \eta_{x_k}$, then $V = \eta$. We get $0 =$

$$\eta \sum_{k=1}^{n} L_{p_k}(Du, u, x)\Big|_{\partial\Omega} - \int_\Omega \eta \sum_{k=1}^{n} (L_{p_k}(Du, u, x))_{x_k} + L_z(Du, u, x)\eta \, dx$$

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By compactness of η we get for all $\eta \in C_0^\infty(\Omega)$

$$0 = \int_\Omega \eta \left(- \sum_{k=1}^{n} (L_{p_k}(Du, u, x))_{x_k} + L_z(Du, u, x) \right) \, dx$$

The First Variation

Therefore, in Ω we have

$$-\sum_{k=1}^{n}\left(L_{p_k}(Du,u,x)\right)_{x_k}+L_z(Du,u,x)=0$$

that is to say that if u is the minimizer to $J[\cdot]$, then it solves the PDE above. This is called the Euler-Lagrange corresponding to our functional $J[\cdot]$.

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If we let $n=1$, then $L(Du,u,x)=f(u',u,x)$ and we get

$$-\frac{d}{dx}\frac{\partial f}{\partial u'}+\frac{\partial f}{\partial u}=0$$

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Recall the chain rule (why does $\partial_x f$ vanish?)

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then

$$u'\frac{\partial f}{\partial u} = \boxed{\frac{df}{dx} - \frac{\partial f}{\partial u'}u'' = u'\frac{d}{dx}\frac{\partial f}{\partial u'}}.$$

Move everything to one side to get

$$0 = \frac{df}{dx} - \left(\frac{\partial f}{\partial u'}u'' + u'\frac{d}{dx}\frac{\partial f}{\partial u'}\right) = \frac{d}{dx}\left(f - u'\frac{\partial f}{\partial u'}\right)$$

Beltrami Identity

Thus we have

$$f - u' \frac{\partial f}{\partial u'} = C$$

a constant. In the next example we will utilize this Identity

Fermat's Principle of Least Time

If velocity of light is given by $\mu = \mu(y)$ continuous, the actual light path connecting $(x_0, y(x_0))$ and $(x_1, y(x_1))$ is the path the extremizes the integral

$$J[y] := \int_{x_0}^{x_1} \frac{\sqrt{1 + (y')^2}}{\mu(y)}\, dx$$

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Observe that this satisfies the Euler-Lagrange equation when $n = 1$.

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Observe that this satisfies the Euler-Lagrange equation when $n = 1$.

In particular since f is explicitly independent of x then it satisfies the Beltrami Identity:

$$f - y' \frac{\partial f}{\partial y'} = C$$

Fermat's Principle of Least Time

Next substitute f in to get

$$\frac{\sqrt{1+(y')^2}}{\mu(y)} - \frac{(y')^2}{\mu(y)\sqrt{1+(y')^2}} = C,$$

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the left hand side becomes

$$\frac{1}{\mu(y)\sqrt{1+(y')^2}} = C \Longrightarrow \frac{dy}{dx} = \frac{\sqrt{1-C^2\mu(y)^2}}{C\mu(y)};$$

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therefore,

$$\frac{C\mu(y)\ dy}{\sqrt{1-C^2\mu(y)^2}} = \ dx \Longrightarrow x = \int \frac{C\mu(y)\ dy}{\sqrt{1-C^2\mu(y)^2}}$$

this integral may/may not be elementary to solve.

References I



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